

Thought Collapse in LLM Agents: A Cross-Task Diagnosis of Capacity Domination in Agentic Reasoning

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Abstract

The ReAct framework prepends an explicit *Thought* step to every agent action, and most agent frameworks treat that Thought as load-bearing. We test the assumption with a three-layer diagnosis on five models (Qwen2.5-14B/32B, DeepSeek-V3, Llama-3.1-70B, and a Llama-3.1-8B boundary cell) and five tasks spanning low to high parametric coverage (GSM8K, MATH-hard, HotpotQA, WebQuestions, TriviaQA). The central finding is **Capacity Domination**: when a model’s capacity exceeds a task’s coverage in pre-training, the Thought step collapses from a useful reasoning step into a redundant or actively harmful one. Behaviorally, the *Thought-Gap* ($EM_{Full} - EM_{None}$) decays from +0.59 on low-coverage GSM8K (14B) to non-significant for 32B on high-coverage WebQuestions and TriviaQA. DeepSeek-V3 retains a small but significant residual gap (+0.047 / +0.067), consistent with a two-stage collapse in which scaffold degradation precedes Thought-level collapse. Cross-family Llama-3.1-70B shows the same decay, with Gap declining monotonically across all five coverage levels and the largest Direct–Full reversal in our matrix (+0.143 EM on TriviaQA, three-seed mean). A *Direct Probe* that strips the entire ReAct scaffold outperforms full ReAct by up to $4\times$ on high-coverage tasks. A Thought Causal Influence (TCI) probe records action–Thought Mismatch of 0.62–0.84 on high-coverage tasks with Follows-Q near 100%: the Action conditions on the question, not on the Thought. Motivated by the diagnosis, we propose *Coverage-Aware Routing* (CAR), which estimates coverage online and routes between Direct and ReAct, recovering up to +0.225 EM while saving 93–99% of Thought tokens. Our analysis is scoped to single-tool, short-horizon agents; whether the same collapse persists under multi-tool, long-horizon composition is an open question.

1 Introduction

The ReAct paradigm (Yao et al., 2023) has become the dominant scaffold for LLM agents. At every step the model emits a free-form *Thought* that reasons about the current state, then issues an *Action* that calls a tool or returns an answer. The implicit assumption is that the Thought is load-bearing: verbalized reasoning that improves the downstream Action. This assumption underlies a large fraction of current agent frameworks, evaluations, and dataset designs.

Recent work has begun to question whether chain-of-thought (CoT) reasoning is causally responsible for the gains it correlates with. Lanham et al. (2023) show that CoTs are often unfaithful: models can reach the same answer regardless of perturbations to the reasoning trace. Sprague et al. (2024) find that CoT delivers most of its benefit on math and symbolic tasks, with negligible gains elsewhere. Mallen et al. (2023) establish that for high-popularity factual queries, parametric memory dominates retrieval. These results suggest that the value of explicit reasoning is task-dependent, and they leave open a more specific question: *when does the agentic Thought step actively hurt, and why?*

The answer, we argue, depends on two quantities: a model’s *capacity* on a task and the task’s *coverage* in pre-training. When coverage is low, parametric memory is unreliable and the Thought scaffold does real work. When coverage is high, the model already has the answer in weights, and Thought tokens become a noise channel that can override correct retrieval rather than refine it. We call this regime **Capacity Domination** and the resulting failure mode **Thought Collapse**.

Three-layer diagnostic protocol. We test this hypothesis with a progressively deeper diagnostic pipeline (Figure 1):

- **Behavioral gap:** the *Thought-Gap* $\Delta_{EM} = EM_{Full} - EM_{None}$, measuring whether removing Thought hurts, helps, or is neutral.
- **Scaffold-level probe:** a *Direct Probe* that strips the entire ReAct scaffold, revealing whether the loop itself is beneficial or harmful.
- **Mechanism probe:** a Thought Causal Influence (TCI) probe that injects controlled Thought perturbations and measures whether the Action uses the Thought.

Contributions.

1. We define Capacity Domination as the regime where model capacity exceeds task coverage, and identify Thought Collapse as its failure mode, a diagnosis distinct from generic CoT redundancy.
2. We provide the first three-layer (behavioral gap $\rightarrow$ scaffold probe $\rightarrow$ mechanism probe) cross-task evidence of Thought Collapse on five tasks and five model scales (Qwen2.5-14B/32B, DeepSeek-V3, cross-family Llama-3.1-70B, and a weak-end Llama-3.1-8B boundary), with three-seed cluster-bootstrap CIs on every primary cell.
3. On high-coverage tasks, removing the entire ReAct loop outperforms full ReAct by up to $4\times$ EM. We call this *scaffold degradation*; it is a stronger effect than CoT redundancy alone.
4. Thought Collapse is a two-stage process. Scaffold degradation comes first (Direct catches up to Full); Thought-level collapse follows (Full stops beating None). This ordering predicts the small residual Thought-Gap on capacity-rescued models like DeepSeek-V3.
5. We propose *Coverage-Aware Routing* (CAR), a lightweight online routing rule that recovers up to $+0.225$ EM while saving 93–99% of Thought tokens, and we release the code, traces, and OpenRouter audit log.

2 Background and Related Work

Chain-of-thought and ReAct. Wei et al. (2022) introduce CoT prompting, showing that intermediate verbal reasoning unlocks compositional and arithmetic abilities at sufficient model scale. Yao et al. (2023) extend CoT to agentic settings by interleaving Thought with tool-using Actions, reporting consistent improvements over Act-only baselines on HotpotQA and ALFWorld. Subsequent agent frameworks largely inherit this Thought-first design. Our **NoThought** condition directly replicates the Act-only baseline of Yao et al. (2023), enabling a head-to-head comparison under identical harness settings.

Faithfulness and necessity of CoT. Lanham et al. (2023) measure CoT *faithfulness* via truncation, mistake injection, and paraphrasing, and find that on many tasks the answer is robust to CoT perturbations, suggesting the verbalized chain does not causally drive the answer. Sprague et al. (2024) meta-analyze 100+ papers and conclude that CoT helps mainly on mathematical and symbolic tasks and is largely neutral elsewhere. Closest to our setting, Verma et al. (2024) run a sensitivity analysis on ReAct in AlfWorld and report that removing the interleaving, or replacing the reasoning trace with placebo guidance, leaves performance largely unchanged; they attribute the apparent gains to exemplar–query similarity rather than reasoning. We reach a compatible conclusion through a different route: instead of a single planning domain and a similarity account, we vary task coverage across five tasks and tie the Thought’s (ir)relevance to a capacity–coverage gap, then turn the diagnosis into a routing rule. Our work extends this picture to the agentic setting: not only is CoT sometimes unhelpful, the entire ReAct scaffold can be *net negative* on high-coverage tasks.

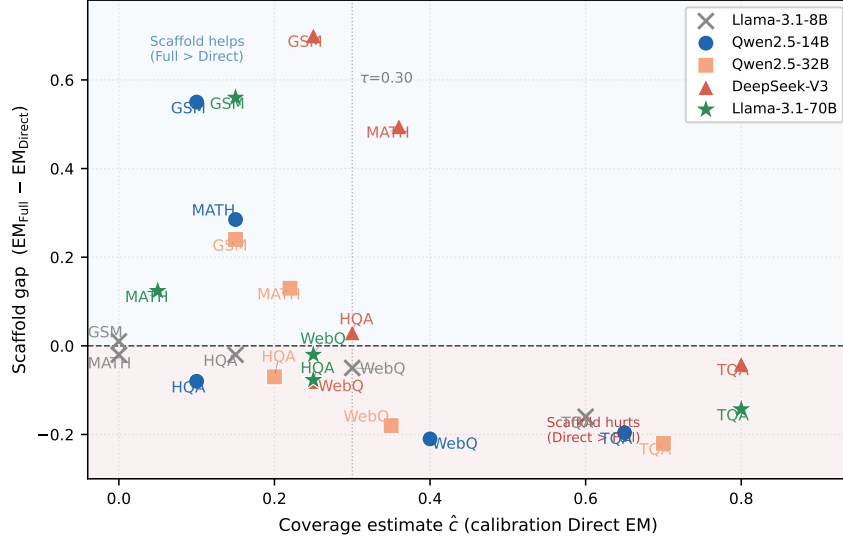


Figure 1: Scaffold gap ($EM_{Full} - EM_{Direct}$) vs. coverage estimate $\hat{c}$ (calibration-set Direct EM) for all 25 (model, task) cells across five models spanning Llama-3.1-8B through DeepSeek-V3. Positive gap = the ReAct scaffold helps over parametric memory alone; negative = the scaffold hurts. The transition from scaffolded to direct-memory regime tracks coverage, confirming the Capacity Domination hypothesis. Marker shape indicates model; task abbreviations: GSM=GSM8K, MATH=MATH-hard, HQA=HotpotQA, WebQ=WebQuestions, TQA=TriviaQA.

Parametric vs. retrieved knowledge. Mallen et al. (2023) introduce a coverage-style notion of *popularity* and show that high-popularity factual questions are answered better from parametric memory than from retrieval. We operationalize coverage as a continuous, model-specific signal ($\hat{c}$, the calibration-set Direct EM) and tie it to a diagnosis of the agentic Thought step, rather than retrieval choice.

Thought Collapse. Our framing is closest to faithfulness work (Lanham et al., 2023), but differs in three ways: (i) we operate in the agentic ReAct loop rather than single-turn QA; (ii) we measure a scaffold-level *Direct Probe* that severs the entire scaffold, not just CoT; and (iii) we tie collapse causally to a coverage-capacity gap and turn it into an actionable routing rule.

Adaptive reasoning routing. Lu et al. (2025) gate per-query reasoning on answer perplexity and report accuracy-efficiency Pareto gains, addressing the same tradeoff between answer accuracy and Thought-token cost. Our work routes at the task level using an external coverage estimate (calibration-set Direct EM) rather than a model-internal uncertainty signal. The two approaches are complementary: task-level routing exploits the regularity that Capacity Domination is stable across the bulk of a task, while a model-internal query-level signal could handle borderline items (Section 8).

Concurrent and contemporary work. Two recent papers bear directly on our claims. Zhou et al. (2026) compare six inference-time paradigms (including Direct and ReAct) across four models and ten benchmarks, find that no fixed paradigm dominates, and train an embedding-based router to pick one per task. Their headline observation, that the same scaffold helps on some tasks and hurts on others, matches ours, and the work is independent and contemporary (their preprint predates ours by a few weeks). We differ in what we provide: rather than a learned router, we give a coverage-grounded account of *why* the scaffold flips sign, supported by the scaffold and mechanism probes, and a routing rule that reads off a 20-example coverage probe with no training. Our task suite is also

organized along a single parametric-coverage axis rather than a heterogeneous benchmark mix, which is what lets the sign flip be read as Capacity Domination rather than task miscellany. Gekhman et al. (2026) study the opposite end of the same axis: on low-coverage closed-book QA, reasoning *expands* parametric recall, with the largest gains on weaker models and the observation that more capable models recall better without reasoning. That monotonicity—reasoning matters most where capacity is low relative to the task—is the same one we report, extended here to the agentic ReAct setting with tool use, where the high-coverage regime makes the inverse effect (Thought Collapse) visible. They further isolate a content-independent *computational buffer* effect, which we return to in Section 9.

3 Experimental Setup

Models. We use three primary open-weight or API-served models spanning a $\sim 20\times$ active-parameter range: Qwen2.5-14B-Instruct, Qwen2.5-32B-Instruct, and DeepSeek-V3 (~ 37 B activated MoE). For cross-family validation we evaluate Llama-3.1-70B-Instruct via OpenRouter, pinned to the DeepInfra bf16 endpoint (UUID prefix 59c87462); a post-hoc audit of 20 sampled calls confirms 100% endpoint compliance, and all generation IDs are released in logs/openrouter_call_audit.jsonl. Llama-3.1-8B-Instruct serves as a weak-end boundary cell (Section 9), pinned to the DeepInfra bf16 endpoint (UUID prefix 858e9b98). All models are queried via cloud APIs with temperature 0. Qwen-14B/32B, DeepSeek-V3, and Llama-3.1-70B carry three-seed replication (seeds 42, 7, 123) on every (model, task) cell across all three probe layers; Llama-3.1-8B is single-seed (42) by design as a boundary check.

Tasks and coverage. Coverage is operationalized as $\hat{c}$, the empirical Direct-probe EM on a 20-example calibration set (the same signal CAR uses in Section 7). Lower $\hat{c}$ means parametric memory alone cannot solve the task; higher $\hat{c}$ means it can. Because $\hat{c}$ is model-specific, we use the range across the three primary models as a task-level descriptor (Table 1). Tasks where $\hat{c} > 0.40$ for most models we label **high-coverage**; below 0.30 we label **low-coverage**; in between is **mid-coverage**. We evaluate five tasks spanning this spectrum:

- **Low coverage** ($\hat{c}$: 0.10–0.36): *GSM8K* (Cobbe et al., 2021) and *MATH-hard* (level-5 subset; Hendrycks et al., 2021).
- **Mid coverage** ($\hat{c}$: 0.10–0.30): *HotpotQA* (Yang et al., 2018).
- **High coverage** ($\hat{c}$: 0.35–0.80): *WebQuestions* (Berant et al., 2013) and *TriviaQA* (Joshi et al., 2017).

Most tasks are evaluated on $n = 100$ examples; WebQ (all models) and TriviaQA-32B/V3 are extended to $n = 200$ for higher-coverage cells where gap estimates were noisier.

Conditions. Three prompt variants share the same ReAct harness: **FullThought** (standard ReAct with Thought:/Action:/Observation:), **NoThought** (the Thought: field is removed and the model must emit Action: directly, equivalent to the Act-only baseline in Yao et al. 2023), and **Direct** (no ReAct scaffold at all, a single zero-shot turn with no tool description).

Tools. The ReAct harness exposes two tools: (i) a **Wikipedia search** that queries the Wikipedia REST API and returns the opening three sentences of the top-ranked article, and (ii) a **calculator** that evaluates arithmetic expressions in a sandboxed Python eval. Factual tasks (WebQ, TriviaQA, HotpotQA) use Wikipedia search; math tasks (GSM8K, MATH-hard) use the calculator. The Direct condition receives no tool description. Our search tool returns *real* retrieved content, which makes the WebQ/TriviaQA result that Direct $\gg$ Full more difficult to dismiss as a retrieval failure.

Metrics. We report Exact-Match (EM) under task-specific normalization, with 95% confidence intervals from a non-parametric bootstrap ($n_{\text{boot}} = 2000$). Significance refers to a CI excluding zero. For multi-seed cells (seeds 42/7/123) we use *cluster bootstrap*: question IDs are resampled with replacement, keeping all seed observations for each resampled question, which accounts for within-question correlation across seeds. Single-seed cells use plain paired bootstrap.

Table 1: Behavioral layer: *Thought-Gap* $\Delta_{EM} = EM_{Full} - EM_{None}$ with 95% bootstrap CIs ($n_{boot} = 2000$). * marks CIs that exclude zero. Per-seed $n = 100$ for most cells; WebQ (all models) and TriviaQA-32B/V3 use $n = 200$ on seed 42 and $n = 100$ on seeds 7/123; MATH-hard reports the larger aggregated n_q (400/264/100). Point estimates and CIs in this table are the seed-42 paired bootstrap; three-seed replication (seeds 42, 7, 123) is available for every Qwen-14B/32B, DeepSeek-V3, and Llama-3.1-70B cell, with across-seed standard error ≤ 0.03 on every cell, addressing the prompt-sensitivity concern raised by Sprague et al. (2024). Coverage label derived from $\hat{c}$ range (in parentheses) across the three primary models; see Section 3.

Task	Coverage ($\hat{c}$)	14B	32B	V3	L70B
GSM8K	Low (.10–.25)	+0.587 [+0.525, +0.649]*	+0.180 [+0.107, +0.251]*	+0.073 [+0.039, +0.110]*	+0.443 [+0.365, +0.520]*
MATH-hard	Low (.15–.36)	+0.385 [+0.335, +0.433]*	+0.106 [+0.030, +0.178]*	+0.180 [+0.080, +0.280]*	+0.077 [+0.013, +0.142]*
HotpotQA	Mid (.10–.30)	+0.103 [+0.069, +0.140]*	+0.070 [+0.026, +0.112]*	+0.120 [+0.070, +0.171]*	+0.010 [−0.043, +0.060]
WebQ	High (.25–.40)	+0.060 [+0.033, +0.089]*	−0.013 [−0.046, +0.021]	+0.047 [+0.019, +0.076]*	−0.020 [−0.054, +0.017]
TriviaQA	High (.65–.80)	+0.300 [+0.244, +0.357]*	+0.020 [−0.030, +0.070]	+0.067 [+0.022, +0.117]*	−0.040 [−0.086, +0.007]

Mechanism probe (TCI). For TCI v2 we sample one Thought-step trajectory per question, perturb the Thought (paraphrase / truncate / replace with question-only / replace with adversarial off-topic), regenerate the Action under each perturbation, and record (i) action–Thought *Mismatch* (the Action ignores the perturbed Thought) and (ii) *Follows-Q* rate (the Action remains semantically aligned with the original question).

4 Behavioral Layer: The Thought-Gap

We first ask whether removing Thought hurts, helps, or is neutral. The Thought-Gap Δ_{EM} is positive when Thought helps and non-positive when it is at best redundant. Table 1 reports Δ_{EM} with bootstrapped 95% CIs on the five tasks.

Findings. Four patterns emerge. (i) On low-coverage tasks ($\hat{c} < 0.30$), all four models show significant Thought-Gaps on *both* GSM8K and MATH-hard (14B: +0.587/+0.385; 32B: +0.180/+0.106; V3: +0.073/+0.180; L70B: +0.443/+0.077). The gap shrinks with model capacity but remains significant, indicating that Thought scaffolding is necessary when task coverage is low regardless of model scale or family. (ii) On mid-coverage HotpotQA ($\hat{c}$: 0.10–0.30), the gap is positive and significant for the three primary models (+0.103–+0.120): multi-hop composition still rewards Thought regardless of scale. (iii) On high-coverage tasks ($\hat{c} > 0.35$), 32B shows non-significant gaps on both WebQ (−0.013, CI [−0.046, +0.021]) and TriviaQA (+0.020, CI [−0.030, +0.070]). This is the typical Capacity Domination case. DeepSeek-V3 retains small but significant gaps (+0.047 on WebQ, +0.067 on TriviaQA), consistent with a *two-stage collapse*: scaffold degradation (Direct $\approx$ Full, see Table 2) precedes complete Thought-level collapse. The data are consistent with a coverage-dependent decay of the Thought-Gap, with 32B at the collapse boundary on high-coverage tasks and V3 just beyond scaffold degradation. (iv) Cross-family Llama-3.1-70B reproduces the entire coverage gradient within a single model: Gap decays strictly monotonically across all five tasks, +0.443* (GSM8K) $\rightarrow$ +0.077* (MATH-hard) $\rightarrow$ +0.010 (HotpotQA, n.s.) $\rightarrow$ −0.020 (WebQ, n.s.) $\rightarrow$ −0.040 (TriviaQA, n.s.; CI upper bound +0.007 touches zero). 70B is the only single model in this study to traverse the coverage spectrum monotonically through zero into negative-Gap territory, indicating that the coverage–Thought relationship is a structural regularity rather than a Qwen/DeepSeek artifact. Section 9 returns to 70B as a within-model natural experiment.

TriviaQA-14B caveat. The 14B model shows a large gap on TriviaQA ($\hat{c} = 0.65$, Gap = +0.300), anomalous given high coverage. Figure 1 resolves this: 14B’s scaffold gap (Full–Direct) is −0.20, meaning the ReAct loop itself is harmful. The behavioral gap is inflated by NoThought’s failure under the scaffold format, not by Thought genuinely helping. Section 5 elaborates.

Table 2: Scaffold-level probe: Direct Probe (no scaffold) vs. FullThought vs. NoThought EM. “Direct $\gg$ Full” indicates scaffold degradation, where the ReAct loop is net harmful; “Direct $\ll$ Full” indicates a genuinely helpful scaffold. Partial scaffold degradation begins at mid-coverage (HotpotQA, 14B/32B). All Qwen-14B/32B, DeepSeek-V3, and Llama-3.1-70B values are three-seed point estimates (seeds 42/7/123, $n = 100$ per seed; WebQ and TriviaQA-32B/V3 use $n = 200$ on seed 42). The L70B $\times$ TriviaQA cell exhibits the largest scaffold reversal across our 25 (model, task) cells (+0.143 Direct–Full EM, three-seed mean, per-seed +0.14/+0.14/+0.15 with $\sigma < 0.01$).

Model	Task	Direct	Full	None	Pattern
14B	GSM8K	0.10	0.65	0.04	Direct $\ll$ Full
32B	GSM8K	0.20	0.44	0.25	Direct $\ll$ Full
V3	GSM8K	0.26	0.96	0.86	Direct $\ll$ Full
L70B	GSM8K	0.183	0.743	0.300	Direct $\ll$ Full
14B	MATH-hard	0.150	0.435	0.090	Direct $\ll$ Full
32B	MATH-hard	0.220	0.350	0.260	Direct $\ll$ Full
V3	MATH-hard	0.360	0.855	0.661	Direct $\ll$ Full (2.4 $\times$)
L70B	MATH-hard	0.163	0.287	0.210	Direct $\ll$ Full
14B	HotpotQA	0.28	0.20	0.07	Direct $>$ Full
32B	HotpotQA	0.28	0.21	0.07	Direct $>$ Full
V3	HotpotQA	0.29	0.32	0.17	Direct $\approx$ Full
L70B	HotpotQA	0.330	0.253	0.243	Direct $>$ Full
14B	WebQ	0.29	0.08	0.01	Direct $\gg$ Full (3$\times$)
32B	WebQ	0.30	0.12	0.15	Direct $\gg$ Full
V3	WebQ	0.22	0.14	0.13	Direct $\gg$ Full
L70B	WebQ	0.200	0.180	0.200	Direct $\approx$ Full
14B	TriviaQA	0.630	0.434	0.138	Direct $>$ Full
32B	TriviaQA	0.660	0.440	0.360	Direct $>$ Full
V3	TriviaQA	0.780	0.739	0.714	Direct $\approx$ Full
L70B	TriviaQA	0.823	0.680	0.720	Direct $\gg$ Full (+0.143)

5 Scaffold-Level Probe: Direct Baseline and Degradation

The behavioral layer compares two points within the ReAct loop. To test whether the loop *itself* is appropriate, we add a third condition: a single-turn Direct prompt with no Thought:, no Action:, and no tool affordance. If ReAct were a strict improvement over zero-shot, Direct should be no better than NoThought. Table 2 shows otherwise.

Three regimes along the coverage gradient. On low-coverage tasks (GSM8K, MATH-hard) Direct is dramatically worse than Full ReAct for every model (e.g., V3 drops from 0.96 to 0.26 on GSM8K). The ReAct scaffold, with calculator-style decomposition, is doing real work. On mid-coverage HotpotQA, a partial reversal already appears: 14B and 32B achieve higher EM with Direct (0.28) than with Full ReAct (0.20/0.21), and V3 closes the gap (0.29 vs. 0.32). On high-coverage tasks (WebQ, TriviaQA), the inversion is complete: *every* model achieves higher EM by skipping the scaffold. The most extreme case is 14B on WebQ, where Direct (0.29) is $\approx 4\times$ Full (0.08); 14B on TriviaQA shows Full erasing 0.20 EM relative to Direct.

Scaffold degradation, not Thought redundancy. This is a stronger statement than “Thought is redundant.” On WebQ-14B, NoThought (0.006) is even worse than Full (0.08): both ReAct variants are below random guessing of the parametric answer (0.29). The ReAct format itself, the act of emitting an Action[search] with arguments, appears to derail parametric retrieval, presumably by biasing the decoder toward query-rewriting tokens rather than answer tokens.

Scale heals partially. V3 on TriviaQA closes the Full–Direct gap (0.780 vs. 0.739): a sufficiently strong model absorbs the scaffold cost. This is the predicted *capacity rescue*, where

Table 3: Mechanism layer (TCI v2). **Mismatch** is the rate at which the regenerated Action departs from the (perturbed) Thought; **Follows-Q%** is the rate at which the Action remains semantically aligned with the original question. Each cell is the three-seed mean (seeds 42, 7, 123) for Qwen-14B/32B, DeepSeek-V3, and Llama-3.1-70B; Llama-3.1-8B is a single-seed boundary cell. Per-seed sampling yields $n \approx 45\text{--}50$ items $\times$ 4 perturbation conditions per cell. Across-seed standard error: Mismatch up to ± 0.046 , Follows-Q within $\pm 3.5\%$; the question-level Follows-Q is the more stable of the two metrics. The L70B $\times$ MATH-hard cell (**0.16, 60%**) is the most Thought-dependent in the matrix; the L8B $\times$ GSM8K cell (Follows-Q 7%) marks the opposite boundary, where the Action is captured by whatever Thought is supplied.

Task	Cov.	Mismatch					Follows-Q%				
		14B	32B	V3	L70B	L8B	14B	32B	V3	L70B	L8B
GSM8K	Low	0.60	0.77	0.44	0.61	0.35	82	91	88	75	7
MATH-hard	Low	0.35	0.47	0.22	0.16	0.30	90	92	95	60	32
HotpotQA	Mid	0.76	0.70	0.71	0.62	0.63	99	100	99	99	86
WebQ	High	0.84	0.76	0.83	0.67	0.63	99	100	99	100	100
TriviaQA	High	0.83	0.76	0.68	0.64	0.63	100	100	100	100	96

capacity offsets collapse on the most-covered task. Smaller models on the same data pay a disproportionate cost.

6 Mechanism Layer: Thought Causal Influence

The behavioral and scaffold-level probes establish that Thought is, at best, idle in the high-coverage regime. The mechanism layer asks *how*: does the Action conditionally depend on the Thought tokens, or is it generated as if the Thought were absent? We measure this with TCI v2: for each item we generate a Thought, then perturb it (paraphrase, truncate, blank, off-topic adversarial) and regenerate the Action under each perturbation. Two metrics summarize the trajectory:

- **Mismatch**: fraction of (item, perturbation) pairs where the regenerated Action departs from the perturbed Thought (i.e., the Action ignored the Thought).
- **Follows-Q%**: fraction of Actions that remain on-topic with the original question across perturbations.

A high Mismatch combined with high Follows-Q is the signature of Thought Collapse: the Action is determined by the question, not the Thought.

Findings. The two metrics give a single coherent picture. On high-coverage tasks (WebQ, TriviaQA) Mismatch is high (0.63–0.84) and Follows-Q sits at or near 100% on every primary model: the Action is conditioned on the question, not on the Thought. On low-coverage MATH-hard the relationship inverts. Mismatch drops to 0.16–0.47 and Follows-Q drops with it (60–95%), meaning a perturbed Thought changes the Action and pulls it off the question, the signature of genuine Thought-conditioning. GSM8K sits between the two regimes (Mismatch 0.35–0.77, Follows-Q 75–91%). Cross-family Llama-3.1-70B reproduces the gradient: its MATH-hard cell (Mismatch 0.16, Follows-Q 60%) is the most Thought-dependent of all 25 cells, and its high-coverage cells (WebQ 0.67/100%, TriviaQA 0.64/100%) fall into the same question-constrained regime as the Qwen and V3 cells. The mechanism-layer signature of Thought Collapse therefore tracks the coverage gradient across four models (three families) and five tasks.

A caveat on factual tasks. On single-entity factual tasks (WebQ, TriviaQA) the gold Action is a named entity already constrained by the question. High Mismatch together with 100% Follows-Q should be read as “the Action is question-constrained, not Thought-conditioned”, the mechanistic claim of Capacity Domination, rather than as evidence of random drift. On reasoning tasks (GSM8K, MATH-hard) the Action space is open-ended—arithmetic expressions or sub-problem decompositions—so the caveat does not apply: Mismatch

Table 4: Coverage-Aware Routing (CAR) results. ΔEM is the per-task accuracy change against Full ReAct; Token Savings is the relative reduction in Thought tokens. For cells routed to Direct, ΔEM is measured on the held-out split ($n = 80$); cells routed to Full ReAct have $\Delta\text{EM} = 0$ by definition. CAR keeps ReAct on low-coverage tasks (GSM8K) and routes high-coverage tasks to Direct, recovering accuracy and most of the token budget.

Model	Task	Coverage Est.	Decision	$\Delta\text{EM vs Full}$	Token Savings
14B	WebQ	0.40	Direct	+0.175	94.8%
32B	WebQ	0.35	Direct	+0.225	96.6%
V3	WebQ	0.25	Full ReAct	0.000	0%
L70B	WebQ	0.25	Full ReAct	0.000	0%
14B	TriviaQA	0.65	Direct	+0.200	96.8%
32B	TriviaQA	0.70	Direct	+0.188	98.9%
V3	TriviaQA	0.80	Direct	+0.000	98.4%
L70B	TriviaQA	0.80	Direct	+0.113	99.2%
14B	GSM8K	0.10	Full ReAct	0.000	0%
32B	GSM8K	0.15	Full ReAct	0.000	0%
V3	GSM8K	0.25	Full ReAct	0.000	0%
L70B	GSM8K	0.15	Full ReAct	0.000	0%

and Follows-Q jointly measure Thought-conditioning, with the L70B $\times$ MATH-hard cell delivering the strongest mechanism-layer signal across the 25 cells.

7 Coverage-Aware Routing

The diagnosis above suggests an immediate intervention: rather than always running ReAct, route based on a task-level estimate of coverage. We propose CAR, a lightweight rule:

1. **Coverage estimate.** On a calibration set of $n_{\text{cal}} = 20$ examples, run the Direct prompt and record the empirical EM: $\hat{c} = \text{EM}_{\text{Direct}}|_{\text{cal}}$. This requires no auxiliary model or extra API calls beyond the Direct probe already run for evaluation.
2. **Routing rule.** Route *all* held-out queries to Direct mode if $\hat{c} > \tau$, otherwise keep Full ReAct. We fix $\tau = 0.30$, which correctly routes all 12 tested (model, task) cells with non-negative ΔEM ; a lower threshold would misroute V3 on low-coverage tasks where parametric coverage is borderline.
3. **Token accounting.** Direct uses only the single-turn zero-shot trace; token savings are reported relative to the Full ReAct budget on the held-out set ($n_{\text{eval}} = 80$).

Findings. At $\tau = 0.30$, CAR achieves non-negative ΔEM on every tested cell. For 14B and 32B, high-coverage tasks route to Direct with gains of +0.175–+0.225 EM and 94–99% token savings. V3 sits near the coverage boundary ($\hat{c} = 0.25$ on both GSM8K and WebQ): with $\tau = 0.30$ it correctly keeps Full ReAct for both, achieving zero loss. This conservatism carries a cost: V3-WebQ has Direct EM = 0.21 > Full EM = 0.13, so a lower threshold would recover +0.08 EM, but would also catastrophically misroute V3-GSM8K ($\Delta = -0.69$ from LOTO validation), since both cells share the same $\hat{c} = 0.25$. A per-query coverage signal (rather than per-task) would resolve this ambiguity; we leave this to future work. On TriviaQA, V3 routes to Direct with near-zero EM cost (simulation $\Delta = +0.000$ on the held-out eval split) while saving 98.4% of tokens; the small residual behavioral gap (+0.067*, Table 1) reflects two-stage collapse where scaffold degradation makes Direct competitive despite Thought retaining some value within the scaffold. On low-coverage GSM8K, all models correctly keep the Full pipeline. Cross-family Llama-3.1-70B reproduces this pattern: $\tau = 0.30$ routes TriviaQA to Direct for a +0.113 EM gain with 99.2% token savings, while correctly keeping Full ReAct on GSM8K, MATH-hard, HotpotQA, and WebQ. The routing rule therefore generalizes outside the Qwen/DeepSeek families. We deliberately use a minimal coverage estimator (20-example calibration EM, no auxiliary classifier) to show the routing signal is tractable even with a near-trivial probe; a more robust estimator would extend the gains to borderline models.

Table 5: Per-query correlation tests across two coverage proxies, two entity rules, and two targets. Each cell reports the count of (model, task) pairs with $p < 0.05$ out of 15. Total: 3/120 nominal hits, indistinguishable from the chance expectation of ≈ 6.0 .

Target	Pageview head	Pageview min	InfiniGram head	InfiniGram min
per-query Direct-EM	1/15	1/15	0/15	0/15
sign(Δ_{EM})	1/15	0/15	0/15	0/15

8 Per-Query Coverage Probes: A Symmetric Null

CAR’s task-level $\hat{c}$ leaves a boundary ambiguity: V3-WebQ and V3-GSM8K both yield $\hat{c} = 0.25$ on calibration but need opposite routing. A per-query signal would resolve this. To avoid the circularity of using model behavior to predict model behavior, we test two external proxies. The result is a robust null.

Setup. The two proxies are Wikipedia pageview (external popularity, via the Wikipedia REST API) and InfiniGram counts (Liu et al., 2024) over Dolma v1.7 (Soldaini et al., 2024) (training-corpus frequency). Both extract question entities through spaCy NER with numerals and stop tokens filtered out, then report $\log_{10}$ of the count. Two entity selection rules are evaluated for each proxy: the most prominent entity (*head*) and the rarest one (*min*). The min rule is motivated by HotpotQA’s annotation guideline of pairing a famous bridge-entity with an obscure target (Yang et al., 2018): when the head entity is misleadingly common, a bottleneck entity should better track question difficulty. Each (proxy, rule) is correlated against two per-query targets: per-question Direct-EM, and sign(Δ_{EM}). With 15 (model, task) cells (three primary models $\times$ five tasks),¹ this yields 120 Pearson tests.

Cross-task: monotonic and concordant with $\hat{c}$. Pageview-head medians follow WebQ 5.04 > TriviaQA 4.23 > MATH-hard 4.01 > HotpotQA 3.99 > GSM8K 3.68. InfiniGram-head produces the same ranking with one anomaly: GSM8K’s median is $\log_{10} = 6.89$, an order of magnitude higher than HotpotQA. The cause is character-name inflation in Dolma. Tokens like *Janet* and *John* appear roughly 10^7 times, and they dominate the bottleneck-entity statistic on synthetic narrative tasks. We treat this as a known artifact of corpus-based proxies. The min rule shifts magnitudes downward but preserves rank order on the other four tasks.

Per-query: a symmetric null. The per-query picture differs sharply. Only 3 of 120 tests reach $p < 0.05$, against a chance expectation of 6.0 under $\alpha = 0.05$. Table 5 groups the count by proxy and target. The three nominal hits are scattered: pageview-head on HotpotQA-32B ($r = -0.318$), pageview-min on TriviaQA-32B ($r = +0.317$), and pageview-head on GSM8K-V3 (this last one on sign(Δ_{EM})). They do not share a sign, a target, or a signal class. InfiniGram, despite being a more direct measure of training exposure, yields zero significant cells across all 60 tests. So the few pageview hits are not replicated by the independent corpus-frequency signal, weakening any causal reading.

Why entity-popularity fails here but works elsewhere. The split is mechanistic, not a measurement issue. Entity-popularity proxies measure how often the model has seen a fact during pre-training. That is the right quantity for triggering retrieval. Min et al. (2025) use almost the same InfiniGram-Dolma signal to fire RAG and report 5–12 EM gains on multi-hop QA. Routing the Thought step asks a different question: whether the query needs step-by-step decomposition, which depends on solution structure rather than entity familiarity. The signal tells the model about *what is in memory*, not about *how to compute the*

¹Per-query n varies by cell (38–99) owing to differing entity-resolution rates and per-query sample sizes across tasks; MATH-hard is aligned to the full 100-item probe set and so carries the largest per-query n , yet still yields no significant correlation, so the null is not an artifact of low statistical power.

answer. This explains why the same proxy works for retrieval gating and fails for reasoning gating.

This null does not rule out per-query routing in general. It only rules out the popularity-based family of signals. Model-internal uncertainty looks more promising: Lu et al. (2025) dynamically gate reasoning on answer perplexity and recover accuracy–efficiency Pareto gains. Question-text features (arithmetic markers, named-entity count, length) are another candidate class. We leave a systematic comparison to future work.

9 Discussion

Two outliers in the Thought-Gap pattern. A naive coverage-only model predicts a strictly decreasing gap with coverage. Two cells require explanation: 14B on TriviaQA (+0.300, large positive) and V3 on MATH-hard (+0.18, larger than 32B). The Direct Probe resolves both. For 14B-TriviaQA, the high gap is not Thought “helping”; it is *NoThought failing harder than FullThought* (0.138 vs. 0.434), while Direct (0.630) outperforms both. The gap is therefore an artifact of scaffold degradation, not of Thought utility. For V3-MATH-hard, capacity is high enough that V3 partially solves problems even from Direct (0.36); the additional gain from Thought (0.855 Full vs. 0.661 None) is real but small relative to GSM8K. Coverage and capacity interact non-linearly, especially on the procedural–factual boundary.

Within-model natural experiments: Qwen2.5-32B and Llama-3.1-70B. Holding the model fixed and varying task coverage exposes Capacity Domination directly. On Qwen2.5-32B, the Thought-Gap collapses to statistical zero on both high-coverage retrieval tasks (WebQ: −0.013, n.s.; TriviaQA: +0.020, n.s.) while remaining significantly positive on the two low-coverage reasoning tasks (GSM8K: +0.180*; MATH-hard: +0.106*, three seeds). Llama-3.1-70B, evaluated as a cross-family check, reinforces this picture by exhibiting strict monotonic decay across all five coverage levels: +0.443* (GSM8K) → +0.077* (MATH-hard) → +0.010 (HotpotQA) → −0.020 (WebQ) → −0.040 (TriviaQA, CI upper bound +0.007). 70B is the only single model in our study to traverse the coverage spectrum monotonically through zero into negative-Gap territory. These two within-model experiments, across distinct model families and capacity scales, rule out the alternative hypothesis that Thought-Gap differences merely reflect model-family idiosyncrasies, and isolate *task coverage* as the operative axis of Capacity Domination.

TriviaQA-14B: scaffold failure, not Thought utility. The 14B anomaly ($\hat{c} = 0.65$, Gap = +0.300) is resolved by the scaffold-level probe: Direct (0.63) $\gg$ Full (0.43), so the gap inflates because NoThought fails harder inside the scaffold than FullThought does. The ReAct loop is the problem, not the Thought step. TCI confirms this: Mismatch is 0.83 for 14B-TriviaQA, and the Action ignores Thought content entirely (Follows-Q 100%). V3 at the same task ($\hat{c} = 0.80$) closes the Full–Direct gap (Direct $\approx$ Full $\approx$ 0.74), the predicted “capacity rescue.” The case is therefore not a counterexample to Capacity Domination, but an instance of stacked scaffold failure: high coverage plus scaffold misalignment.

Two-stage collapse and the V3 residual gap. DeepSeek-V3 retains small but statistically significant Thought-Gaps on high-coverage tasks (+0.047 on WebQ, +0.067 on TriviaQA) despite scaffold degradation (Direct $\approx$ Full $\approx$ 0.74 on TriviaQA). This is consistent with a *two-stage collapse* sequence: scaffold degradation (Direct catches up to Full) precedes Thought-level collapse (Full ceases to beat None). V3 has completed stage one but not stage two on TriviaQA: Thought still adds value within the scaffold, but the scaffold itself has already become a liability relative to Direct. 32B, by contrast, shows non-significant gaps on both WebQ and TriviaQA. Stage two is complete here, the typical Capacity Domination case. Llama-3.1-70B on TriviaQA pushes stage-one scaffold degradation to its observed maximum: Direct EM (0.823) exceeds Full EM (0.680) by +0.143 across three seeds (per-seed +0.14/+0.14/+0.15, $\sigma < 0.01$), and its Thought-Gap (−0.040) crosses into the predicted post-stage-two regime, giving the strongest joint scaffold-and-Thought collapse cell in our matrix.

Boundary on the weak-model side: Llama-3.1-8B reproduces Lanham. At the opposite end of the capacity spectrum, we ran the same five-task protocol on Llama-3.1-8B as a boundary check (single seed, $n = 100$). On GSM8K, 8B exhibits a *negative* Thought-Gap (Full = 0.07 < None = 0.15, $\Delta = -0.08$): the model performs strictly worse with explicit Thoughts. On TriviaQA, Direct (0.61) outperforms Full (0.45) by +0.16 EM. The mechanism layer explains why. On GSM8K the TCI probe records Mismatch 0.35, Follows-Q only 7.1%, and similarity-to-misdirection 0.875: when the original Thought is swapped for an adversarial off-topic substitute, the Action does not fall back to the question—it follows the substitute. MATH-hard reproduces the capture pattern (Follows-Q 31.7%, similarity-to-misdirection 0.506), and the effect attenuates on higher-coverage tasks (Follows-Q $\geq 86\%$ on HotpotQA, WebQ, TriviaQA) where the question alone is a strong enough anchor. This is the mechanism-layer counterpart of Lanham’s finding for small models: the ReAct format imposes a cost the 8B model cannot pay, and it leaves the Action steerable by whatever Thought is supplied, including adversarial ones. Together with the cross-family 70B results, this delimits the operational regime of Capacity Domination. The actionable thought-needed regime, exemplified by Qwen-14B/32B and DeepSeek-V3 on low-coverage tasks, sits between two distinct collapse modes: a Lanham-style left-end collapse where the scaffold cost exceeds available capacity, and the right-end Capacity-Domination collapse where capacity exceeds task coverage.

Is the gap compute or content? Gekhman et al. (2026) show that reasoning tokens can help purely as a *computational buffer*: even semantically empty filler, repeated to the original trace length, lifts low-coverage recall. This raises a fair question for our high-coverage cells: does Direct beat Full because the Thought content is useless, or because the extra Thought tokens are actively in the way? Our mechanism layer speaks to this without a new experiment. The buffer effect of Gekhman et al. (2026) is a low-coverage phenomenon: it surfaces latent knowledge that is hard to reach in a single forward pass. In our high-coverage regime the answer is already reachable from Direct (e.g. V3 and 70B exceed 0.74 and 0.82 EM on TriviaQA with no scaffold at all), so there is little latent knowledge for extra computation to recover. The TCI probe is also content-targeted rather than length-targeted: replacing the Thought with a question-only or adversarial off-topic substitute, holding length roughly fixed, changes the Action on high-coverage tasks (Mismatch 0.62–0.84) while the Action stays aligned with the question (Follows-Q near 100%). If the gap were a pure length effect, content substitution at matched length should not move the Action. The two accounts are therefore not in tension: Gekhman et al. (2026) characterize where added computation *recovers* knowledge (low coverage), and we characterize where the scaffold’s content *interferes* with knowledge already in weights (high coverage). A fully length-controlled buffer ablation in the agentic setting would sharpen the separation and is left to future work.

Implications for agent design. Our results argue for two changes in default agent practice. First, *always include a Direct baseline*: it is cheap and frequently dominates ReAct. Second, *condition the Thought budget on a per-task coverage estimate* rather than amortize it across coverage classes; CAR shows this is tractable with a 20-example probe. A related methodological implication is that aggregate evaluation across coverage classes hides the collapses a coverage-stratified evaluation exposes.

Statistical scope. Our claims rest on patterns across 25 (model, task) cells—five tasks $\times$ five models—rather than on individual point estimates. The three primary models (Qwen-14B/32B, DeepSeek-V3) and the cross-family Llama-3.1-70B carry three-seed replication (seeds 42, 7, 123) on every cell across all three probe layers; statistics use cluster bootstrap resampling on question IDs to account for within-question correlation across seeds, addressing the prompt-sensitivity concern raised by Sprague et al. (2024). Llama-3.1-8B is single-seed by design as a weak-end boundary cell. Per-seed $n = 100$ throughout, except WebQ and TriviaQA-32B/V3 which extend to $n = 200$ on seed 42.

Coverage signal granularity. The estimate $\hat{c}$ aggregates at the (model, task) level. Section 8 tests two per-query proxies (Wikipedia pageview and InfiniGram over Dolma) and finds

a symmetric null: entity-popularity is outside the operational scope of reasoning routing. Lifting CAR to per-query therefore needs a different signal class. Model-internal uncertainty (e.g., answer perplexity, Lu et al., 2025) and solution-structure features remain promising candidates that could disambiguate borderline cells like V3-WebQ vs. V3-GSM8K (both at $\hat{c} = 0.25$).

Routing threshold validation. Threshold $\tau = 0.30$ was validated post-hoc by leave-one-task-out (LOTO) cross-validation. For 14B and 32B, LOTO selects $\tau^* = 0.15$ on every held-out task, matching the eval-set sweep. For V3, LOTO selects $\tau^* = 0.15$ on every task except GSM8K, where it would cause catastrophic mis-routing ($\Delta = -0.688$); we therefore set $\tau = 0.30$ as a safety margin. Llama-3.1-70B produces $\tau^* \in \{0.15, 0.25\}$ across the five LOTO folds, fully within the Qwen and V3 range; the WebQ held-out fold shows a mild $\Delta = -0.013$ misroute, an order of magnitude smaller than V3’s GSM8K case. This rules out a Qwen/DeepSeek training-distribution dependency. A held-out split or richer validation protocol would still be more rigorous in deployment.

Beyond single-tool agents. Our harness uses two tools (Wikipedia search and a calculator), one per task family. Whether Thought Collapse generalizes to long-horizon, multi-tool agents—where tool composition itself may rely on verbalized planning—remains an open question, and a natural follow-up to this work.

10 Conclusion

We diagnose Thought Collapse in ReAct agents at three layers: behavior (*Thought-Gap*), scaffold (Direct Probe), and mechanism (TCI). Across five models (Qwen2.5-14B/32B, DeepSeek-V3, Llama-3.1-70B, and a weak-end Llama-3.1-8B boundary) and five tasks spanning low to high coverage, the data support one coverage-driven story: when capacity exceeds coverage, the Thought step decouples from the Action and the entire ReAct scaffold degrades parametric retrieval. Llama-3.1-70B reproduces the gradient across families, with strict monotonic Gap decay over all five coverage levels and a +0.143 Direct–Full EM reversal on TriviaQA. Llama-3.1-8B marks the opposite boundary: at insufficient capacity, the Action is captured by whatever Thought is supplied, including adversarial off-topic ones. CAR, the coverage-aware routing rule we derive from the diagnosis, recovers up to +0.225 EM and saves 93–99% of Thought tokens. The recommendation is therefore narrow: include a Direct baseline before committing to ReAct, and condition the Thought budget on a small-calibration coverage probe. Code, traces, and the OpenRouter audit log are released for replication.

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